The Structure of Information Networks **Sample Solutions to Homework 2**
CS/INFO 6850 Fall 2024

(1a) For E_j to happen, each of the k sensors independently has to miss the interval $[j-1, j]$. Therefore $\Pr[E_j] = (1 - \frac{1}{n})^k$.

(1b) We show the contrapositive. If G is not connected, then there exist adjacent nodes v, w on the segment at distance greater than 2. But then there is an interval of the form $[j, j+1]$ between them with no nodes in it.

(1c) By (b), it is enough to show that $\Pr[\cup_j E_j]$ goes to 0. By the Union Bound, we have $\Pr[\cup_j E_j] \leq n(1 - \frac{1}{n})^k$. So if we choose $k = 2n \ln n$, then

$$\Pr[\cup_j E_j] \leq n(1 - \frac{1}{n})^{2n \ln n} \leq ne^{-2 \ln n} = n \cdot n^{-2} = n^{-1}.$$

(2a) For $j < i$, we define

$$Y_{ij} = \begin{cases} 1 & \text{node } v_i \text{ links to node } v_j \\ 0 & \text{otherwise} \end{cases}$$

Let Z_j be the in-degree of node v_j; we have $Z_j = \sum_{i=j+1}^{n} Y_{ij}$.

(2b) Note that $E[Y_{ij}] = \Pr[Y_{ij} = 1] = \dfrac{1}{i-1}$, since node v_i is choosing where to link uniformly from among the first $i-1$ nodes. Using the fact from (2a) that $Z_j = \sum_{i=j+1}^{n} Y_{ij}$, linearity of expectation tells us that

$$\begin{aligned} E[Z_j] &= \sum_{i=j+1}^{n} E[Y_{ij}] \\ &= \sum_{i=j+1}^{n} \frac{1}{i-1} \\ &= \sum_{k=j}^{n-j} \frac{1}{k} \\ &= H_{n-1} - H_{j-1}. \end{aligned}$$

(2c) Consider a node v_j, every node v_k with $k > j$ has probability $1 - \frac{1}{k-1}$ not to link to v_j. So if we have random variable X_j s.t.

$$X_j = \begin{cases} 1 & \text{node } v_j \text{ has no in-coming links} \\ 0 & \text{otherwise} \end{cases}$$

then

$$\begin{aligned}\mathrm{Exp}[X_j] &= Pr[\text{no nodes links to } v_j] \\ &= \prod_{k=j+1}^{n}\left(1-\frac{1}{k-1}\right) \\ &= \frac{j-1}{j}\cdot\frac{j}{j+1}\cdot\frac{j+1}{j+2}\cdots\frac{n-2}{n-1} \\ &= \frac{j-1}{n-1}\end{aligned}$$

(2d) The total number of nodes with no in-coming links is the random variable $X = \sum_{j=1}^{n} X_j$. Therefore, by linearity of expectation, the expected number of nodes without in-coming links is

$$\mathrm{Exp}[X] = \sum_{j=1}^{n}\mathrm{Exp}[X_j] = \sum_{j=1}^{n}\frac{j-1}{n-1} = \frac{1}{n-1}\sum_{j=1}^{n}(j-1) = \frac{1}{n-1}\cdot\frac{n(n-1)}{2} = \frac{n}{2}$$

(3a) Recall that $k = \log n$. For a single tournament slot in round j, there are 2^j contestants who could occupy this slot, and all equally likely to do so. Therefore, the probability that the person's guess for this precise slot is correct is 2^{-h}.

The number of slots in round j is $n/2^j = 2^k/2^j = 2^{k-j}$. Therefore, the probability that all the person's guesses in round j are incorrect is

$$(1-2^{-j})^{2^{k-j}}.$$

Taking the complementary event, the probability that at least one guess at round j is correct is

$$p(j) = 1-(1-2^{-j})^{2^{k-j}}.$$

(3b) If we write $h = \alpha\log n$ and $k-h = (1-\alpha)\log n$, then

$$\begin{aligned}1-p(h) &= \left(1-2^{-\alpha\log n}\right)^{2^{(1-\alpha)\log n}} \\ &= \left(1-n^{-\alpha}\right)^{n^{1-\alpha}} \\ &= \left[\left(1-n^{-\alpha}\right)^{n^{\alpha}}\right]^{n^{1-2\alpha}}\end{aligned}$$

Looking at this last expression, we recall that for $n^\alpha \geq 1$, the quantity $(1-n^{-\alpha})^{n^\alpha}$ is between $1/4$ and $1/e$. Therefore,

$$4^{-n^{1-2\alpha}} \leq 1-p(h) \leq e^{-n^{1-2\alpha}}.$$

Now, when $\alpha < 1/2$, both of the outer expressions are going to 0 with n (since the exponent is going to $-\infty$), while when $\alpha > 1/2$, both of the outer expressions are going to 1 with n (since the exponent is going to 0). Hence $\alpha^* = 1/2$ is a critical value of α according to the definition in the question.